Probabilistic Machine Learning

Lecture 01: Introduction

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Why Probability?

Guiding Question: Two models both say "fraud". What would you need to know before acting on either — and why does a label not tell you?

Machine Learning Gives Us Predictions

Three are flagged as fraud

and the three labels are identical

Café Central €4.20 legitimate	Supermarket €63.10 legitimate	**Electronics Online** €1,299.00 **fraud**	Petrol Station €78.40 legitimate
Gift Cards ×12 €600.00 **fraud**	Bookshop €24.90 legitimate	**Hotel Booking** €410.00 **fraud**	Pharmacy €12.75 legitimate

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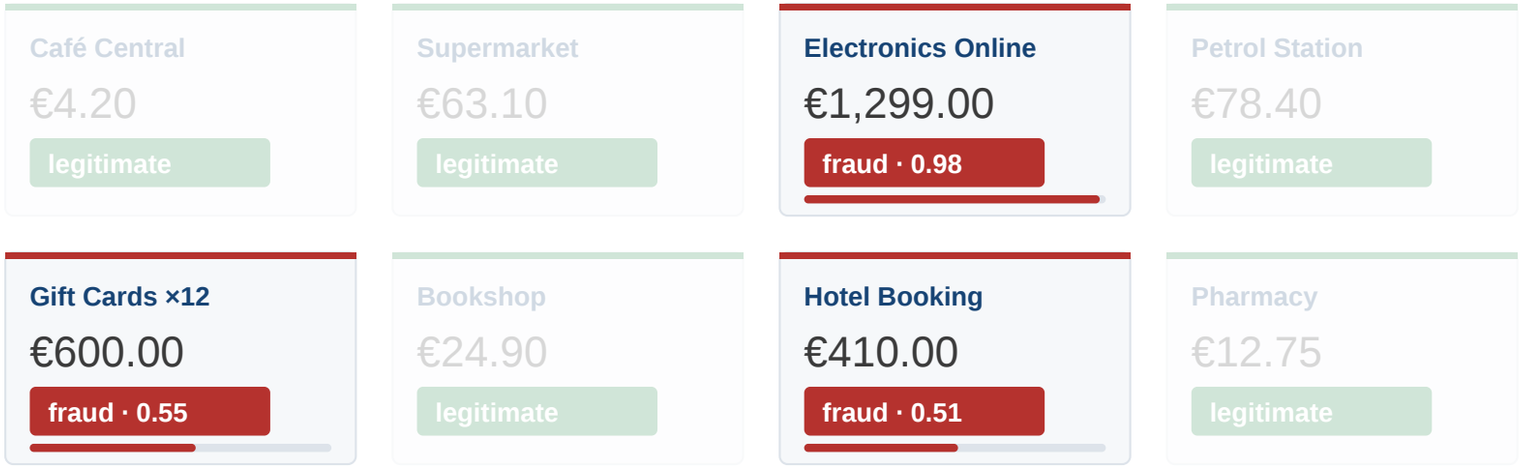
What would you need to know to treat them differently?



The Same Batch, With a Number Attached

Now every flagged case carries a number

0.98, 0.55, 0.51 — three very different cases



What can you do now that you could not do a minute ago?

What a Number Lets Us Do

The number lets us	What that means for this batch
Compare and rank	separate the €1,299 order from the two borderline cases — and compare two models that would both just say "fraud"
Weigh consequences	set the threshold from the cost of a missed fraud against the cost of an annoyed customer
Defer deliberately	send 0.55 and 0.51 to a human; ask whether more data would move them
Combine and communicate	merge evidence from several signals; report *how sure* we are, not only what we think

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> **We need to model uncertainty** — and we need to be able to **reason with it**.

From Prediction to Managed Uncertainty

Predict
a label or a value

→

Quantify
how sure we are

→

Model
represent it explicitly

→

Reason
combine, compare, update

→

Manage
act, defer, or gather data

And once we want to manage it — where does the uncertainty actually come from?

> Probabilistic machine learning is not a different way of producing predictions. It is a **language for reasoning about predictions that are not certain**.

Mini-Exercise

A loan applicant has a **default probability of 0.2**. A default costs the bank **€9,000**; rejecting a customer who would have paid costs **€1,000** in lost margin.

Accept or reject?

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Accepting costs $0.2 \cdot 9{,}000 = €1{,}800$ in expectation; rejecting costs $0.8 \cdot 1{,}000 = €800$. **Reject —** although p is far below 0.5.

What the Threshold Tells Us

THE INTUITIVE RULE

reject if $p > 0.5$

accepts this applicant — whose default risk is **twice** the bank's break-even

THE RULE THE COSTS IMPLY

reject if $p > 0.1$

because a default costs **9×** what a lost customer does: $p^* = \frac{1{,}000}{1{,}000+9{,}000}$

The threshold comes from the **costs**, not from the probability. 0.5 is only the right cut-off when both mistakes are equally expensive — which they almost never are.

Takeaway

A prediction without a measure of its uncertainty cannot be compared, weighed against consequences, or handed to a human at the right moment. **Modelling uncertainty and reasoning with it** is what turns a prediction into a decision.

All of this assumes the numbers are **calibrated**: a score of 0.55 is right about 55% of the time. Whether a model's output deserves that reading is a question Lecture 04 answers.

Sources of Uncertainty

Guiding Question: The model says 0.55. Is that because the world is noisy, or because the model has not learned enough — and would more data help?

Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

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Is the transaction itself ambiguous — would a perfect model, with all the data in the world, still be unsure?

Or has the model simply not seen enough cases like this one?

Would collecting more data help — and could *any* amount of data remove the doubt entirely?

Two Examples

EXAMPLE A · AN AMBIGUOUS OBSERVATION

Two patients with **identical** records — same age, same symptoms, same test results. One recovers, one does not.

✗ Would more patients like these help? Never.

EXAMPLE B · AN UNFAMILIAR CASE

A house-price model trained on flats of 40–120 m² is asked about a **600 m² villa**. It has no examples anywhere near it.

✓ Would ten villas in the training data help? Completely.

Which uncertainty can be reduced, and which is part of the problem itself?

Aleatoric vs. Epistemic Uncertainty

	Aleatoric	Epistemic
Name	*alea*, dice — noise in the world	*episteme*, knowledge — ignorance in the model
Where it comes from	the outcome genuinely varies given what we observe	we have not seen enough to pin the model down
More data?	✗ never — irreducible	✓ yes — reducible
Our example	identical patients, different outcomes (A)	the 600 m² villa (B)

Why the Distinction Matters

If the uncertainty is…	The right response	The expensive mistake
aleatoric (in the problem)	model the noise; set a policy for the ambiguous cases	buying more data that cannot help
epistemic (in our knowledge)	collect data where the model is thin; defer to a human who knows more	accepting it as inherent and never improving

A model that can tell the two apart can report that it is **out of its depth**, not merely unsure.

> A probabilistic model can represent **both** kinds of uncertainty. A Bayesian one can even report them separately — noise and ignorance as two terms of one prediction (Lecture 07). That is why the distinction is worth a name.

Takeaway

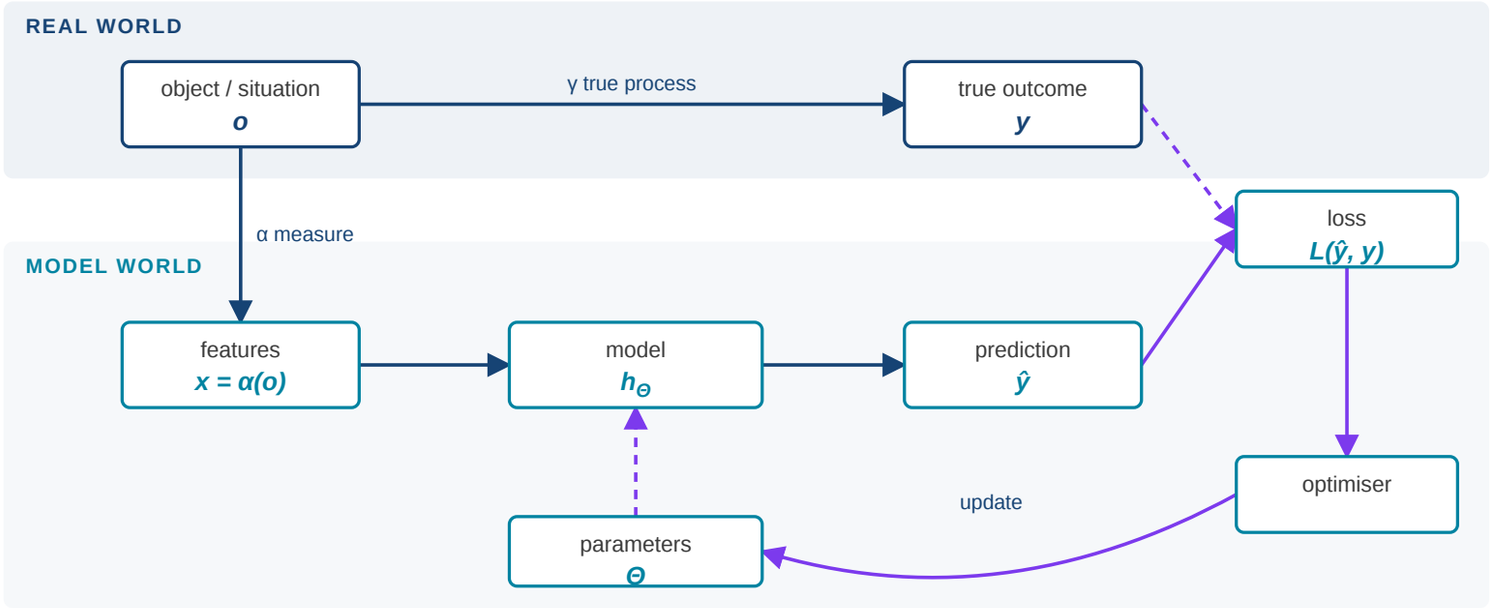
> Uncertainty has two sources: **noise in the world** and **gaps in our knowledge**. Only the second shrinks with data, and a decision — collect more, defer, act — depends on knowing which one you face.

Uncertainty in the Learning System

Guiding Question: You have built learning systems before. Which of their parts were uncertain — and which of those did your model ever report?

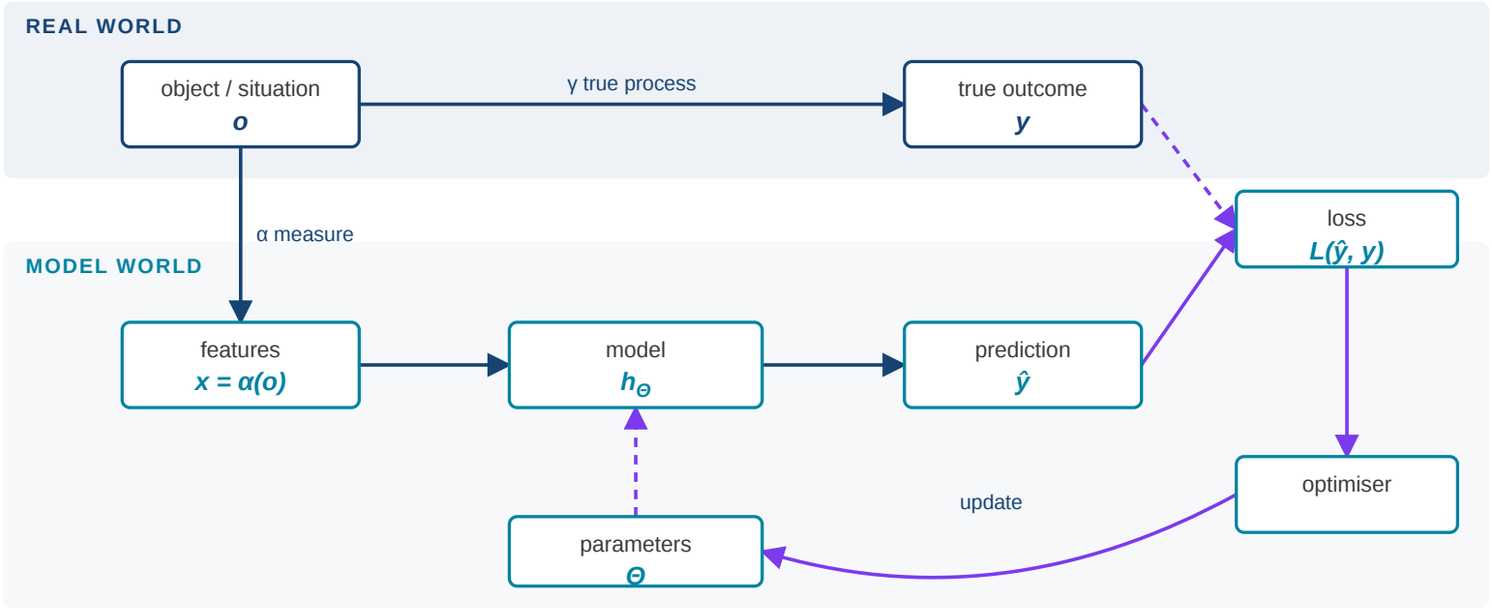
The Learning System You Already Know

The optimiser changes the parameters — and the training loop closes



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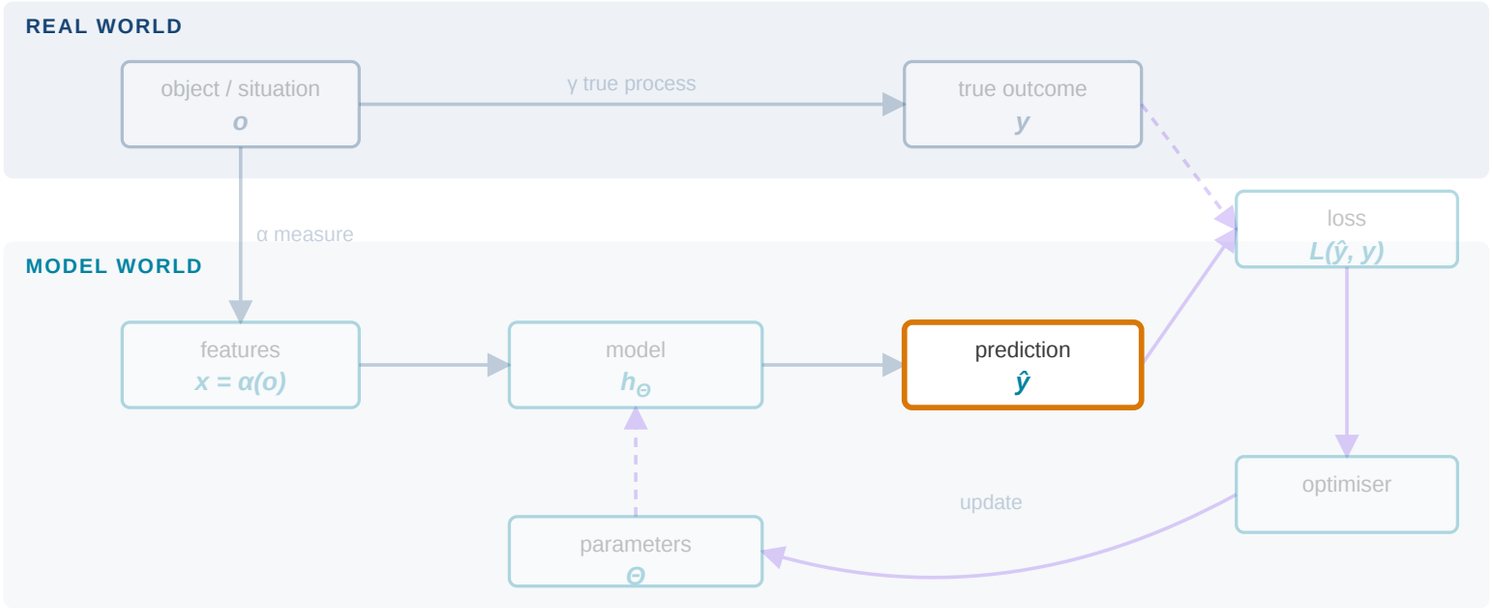
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Which of these boxes did any model you have trained ever report its uncertainty about?

The Same System, Read for Uncertainty

In the evaluation



a finite test set, and a future that may not resemble the past

Uncertainty in vs. about the Loss

	What is uncertain	Kind	More data?
In the loss	the targets y are noisy	aleatoric	✗ never
	the training sample is finite	epistemic	✓ yes
About the loss	whether this objective represents the task at all	epistemic, but a modelling judgement	✗ only knowledge of the task

> Do not say "the loss is random". The loss is an ordinary function — say what is uncertain: the data it is computed from, or whether it is the right objective.

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Uncertainty is not a property of the final prediction. It **enters throughout the system** — and probabilistic machine learning is the language for representing and reasoning about it at every one of these places.

Reasoning Under Uncertainty

Guiding Question: A proof either holds or it does not. A learned prediction is never a proof — so what kind of conclusion is it, and how strong is it?

Three Modes of Reasoning

	From	To	What it guarantees
Deduction	premises	what **must** follow	certain — but adds no new content
Induction	observed patterns	a **regularity**, a hypothesis	nothing — content-adding, not guaranteed
Abduction	an observation	the **best available explanation**	nothing — only comparable against rivals

Machine learning lives in the bottom two rows.

The Cycle, on a Real Case: Uranus

DEDUCTION

Newton's laws plus the known planets give the orbit Uranus **must** follow.

INDUCTION

Six decades of observed positions show a **systematic** deviation — larger than any measurement error.

ABDUCTION

Best explanation: an **unseen planet** further out, pulling on Uranus.

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The cycle turns: the abduced planet becomes a new premise. Le Verrier deduces where it must be; in 1846 Galle points a telescope there and finds **Neptune**, about a degree from the prediction.

An Anomaly Leaves Two Moves

SAVE THE THEORY

Add a hidden cause

The bet: the theory is right; our picture of the world is incomplete.

If wrong: we chase something that is not there.

CHANGE THE THEORY

Replace the premises

The bet: the theory fails here, however well it worked elsewhere.

If wrong: we throw away two centuries of correct predictions.

1859: Mercury's orbit turns faster than Newton allows — about 43″ per century by the modern figure. Which move would you make?

Same Move, Opposite Outcomes

anomaly	the move	what happened
Uranus, 1840s	a hidden planet	**Neptune** found, 1846 ✓
Mercury, 1859	a hidden planet, "Vulcan" (Le Verrier again)	?

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Vulcan is **never found** ✗ In 1915 **general relativity** explains the 43″ with no new planet — here, changing the theory was right.

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> Both decisions were **reasonable when they were made**. What separated them was evidence that did not exist yet.

Deciding Before the Evidence Is In

- → **Strength of belief.** After two centuries of success, absorbing one anomaly *into* the theory was the rational first move.
- → **Evidence accumulates.** Every failed search made "a hidden planet" less plausible — belief moved step by step, not in one jump.
- → **Replace, don't just refute.** Newton fell when a rival explained *more*, not at the first failure.
- → **Costs decide.** Chasing Vulcan forever and abandoning a good theory too early are both expensive; which move to make depends on what being wrong costs.

> Beliefs of different strength, updated by evidence, acted on under costs — the whole lecture in one historical case. Lecture 03 gives each of these words its mathematics.

Mini-Exercise — Your Model Fails on a Subgroup

A fraud model misses most fraud from a **newly added country**. You add 500 more labelled transactions from there, retrain — and the error **does not move**.

Noise, too little data, or the wrong model?

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Noise, too little data, or the wrong model?

Not too little data — then the error would have dropped. Left: the wrong model (try new features or a richer model) or noise (nothing helps, and experts with the same information cannot tell either). You saved the theory once; now the evidence says to change it.

Takeaway

> Learning from data never proves; it **bets**. An anomaly can be absorbed or can overturn the model, and which move is right is decided **under uncertainty** — by how strong the belief is, how the evidence accumulates, and what being wrong costs.

Beyond True or False

	Classical reasoning	Probabilistic reasoning
Conclusions	follow, or do not	are supported to *degrees*
Evidence	premises, taken as given	may be partial or noisy
Certainty	the conclusion is certain	the conclusion may stay uncertain
New information	can only add premises	can *change* what we believe

Classical logic asks *whether* a conclusion follows. Probabilistic reasoning also asks **how strongly the evidence supports each possible conclusion**.

What the Language Has to Do

We now know that uncertainty matters and that it has different sources. What we still lack is a **systematic way** to:

1. **Represent** — say how much we believe each possibility
2. **Combine** — put several uncertain pieces of information together
3. **Update** — change our beliefs when new evidence arrives
4. **Decide** — act on beliefs that are not certainties

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> **Probability provides that language.** For representing, combining and updating there is essentially one consistent set of rules; deciding adds a second ingredient — what the outcomes cost.

Takeaway

Machine-learning predictions are inferences from evidence, not derivations from premises. **Probability is the calculus of such inferences** — which is why Lecture 02 is about it, and why it will feel like an answer rather than a prerequisite.

What Probabilistic ML Enables

Guiding Question: Which questions about a machine-learning system will you be able to answer after this course that you cannot answer today?

After This Course, You Will Be Able to Ask — and Answer

ABOUT A PREDICTION

- ? How certain is it, and should I trust it?
- ? When should the model refuse to decide?
- ? What alternatives are plausible?

ABOUT THE DATA

- ? When would more data help?
- ? When is uncertainty unavoidable?
- ? What does this data actually justify?

ABOUT LEARNING

- ? How should belief change when evidence arrives?
- ? How do I combine prior knowledge with observations?
- ? How do I compare competing explanations?

ABOUT THE METHODS

- ? Why do the common methods work, and under which assumptions?
- ? When should I expect them to fail?
- ? What is my model assuming without telling me?

What Changes Compared to Traditional Machine Learning

TRADITIONAL ML ASKS

What is the prediction?

PROBABILISTIC ML ALSO ASKS

- ? **How certain** is it?
- ? **What alternatives** are plausible?
- ? **Why** are we uncertain?
- ? **How does new evidence** change what we believe?
- ? **What decision** follows from this?
- ? **What can and cannot** be inferred from this information?

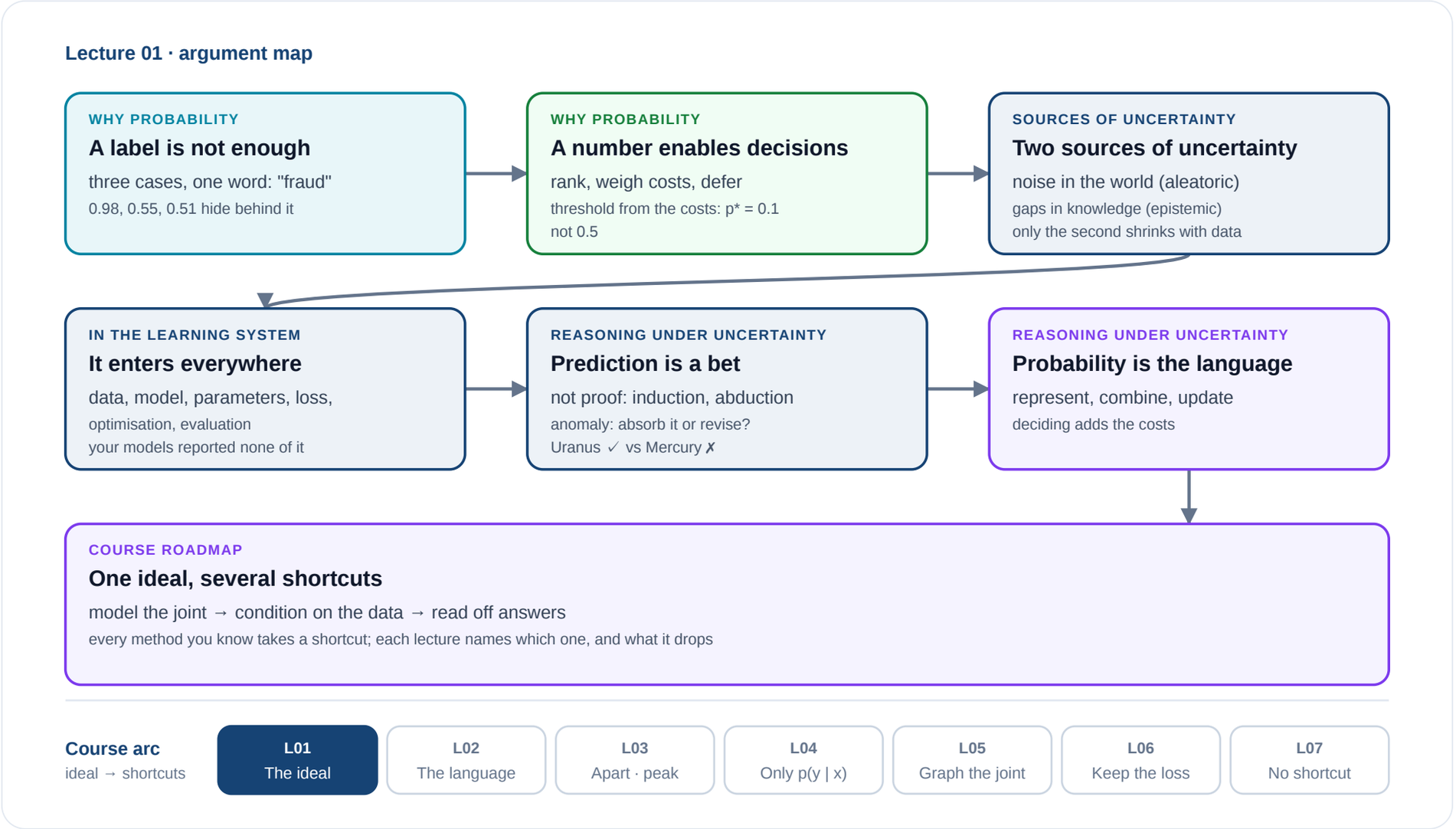
Takeaway

> Probabilistic machine learning does not replace prediction. It **surrounds every prediction with the questions a decision needs answered** — and gives one language in which all of them can be asked.

Course Roadmap

Guiding Question: Now that the motivation is in place — which lecture answers which of your questions?

Argument Map



The Road Ahead

Each lecture answers one of the questions you have just met.

02

Representing uncertainty *How do we describe uncertainty mathematically?*

Probability, distributions, and how to summarise data.

03

Generative models and naive Bayes *What does a classifier look like when it is built straight from Bayes — and what does it cost?*

Keeping the peak, modelling the joint $p(x, y)$, naive Bayes.

04

Discriminative models *What do linear and logistic regression assume — and when is their output a probability?*

Modelling only $p(y \mid x)$: linear and logistic regression, calibration.

05

Structured models *How do we write down a large, structured set of beliefs?*

Graphical models: drawing the assumptions, fitting them, computing with them.

06

Judging models *What is the quantity we actually want to minimise — and what did we get instead?*

Evaluation and empirical risk minimisation.

07

Deciding and refining *How do we keep uncertainty all the way to the decision?*

Mixtures, Bayesian regression, and what comes after.

The Ideal, in Three Moves

Every method in this course is measured against the same three moves, in the language you are about to learn:

Model the joint

Write down what the world would have to be like to produce what we observed — the data *and* the unknowns together.

Condition on the data

Inference is conditioning: fix what was observed, and let the rules of probability say what follows for the rest.

Read off answers

Predictions, uncertainties and decisions are all quantities computed from that one conditioned model.

> Every one of these words — joint, condition, inference — gets a precise meaning in Lecture 02. They appear here only so that you recognise them when they return.

…and the Shortcuts Everyone Takes

Most methods you already know take a shortcut — and each one drops something.

Shortcut	Instead	What is lost	L
Keep the peak	a single best parameter (maximum likelihood or MAP)	how sure we are about it	03–04
Assume it apart	factorise the joint by independence	the dependences assumed away	03
Only what you need	model $p(y \mid x)$, never the joint (discriminative)	every other question	04
Graph the joint	state exactly which independences hold	cheap inference, in general	05
Keep the loss	minimise average error on the sample (ERM)	every probability	06
No shortcut	keep the whole posterior — EM only for the hidden labels	computation; it cannot see a wrong model family	07

> Knowing which shortcut a method takes tells you which questions it can no longer answer.